THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The chemical bond formed by the sharing of electrons is: 1

(a) no bond **(b)** covalent bond
(c) electrovalent bond **(d)** ionic bond

2. According to Dalton, atoms combine in: 1

(a) any ratio **(b)** small whole-number ratios
(c) decimal ratios **(d)** fractional ratios

3. The formula unit mass of $MgCl_2$ is (Mg = 24, Cl = 35.5): 1

(a) 71 u **(b)** 83.5 u **(c)** 95 u **(d)** 59.5 u

4. Which of the following is an ionic compound? 1

(a) $NaCl$ **(b)** CO_2 **(c)** H_2O **(d)** CH_4

5. A molecule is best described as: 1

(a) a mixture of atoms
(b) two or more atoms chemically bonded together that can exist independently
(c) a single atom only
(d) a charged atom

6. An ion is: 1

(a) a neutral atom **(b)** a mixture
(c) a charged atom or group of atoms **(d)** a molecule only

7. Which of the following is a covalent compound? 1

(a) KBr **(b)** $NaCl$ **(c)** CH_4 **(d)** MgO

8. The atomicity of ozone is: 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** 2

9. A cation is a: 1

(a) positively charged ion **(b)** molecule
(c) negatively charged ion **(d)** neutral particle

10. Ionic compounds in the molten or dissolved state: 1

(a) are gases **(b)** are liquids at room temperature always
(c) do not conduct **(d)** conduct electricity

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the formula of
(a) sodium oxide
(b) calcium oxide. — 2

12. Define a molecule. Give one example each of the molecule of an element and of a compound. — 2

13. What is an ion? Differentiate between a cation and an anion. — 2

14. Why is the law of constant proportions not applicable to mixtures? — 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the formulae of
(a) calcium chloride
(b) aluminium oxide
(c) magnesium nitride. — 3

16. Write the names of the ions: Na^+, Mg^{2+}, Cl^-, O^{2-}, OH^-, NH_4^+. — 3

17. What are polyatomic ions? Give three examples with their charges. — 3

OR

Calculate the molecular masses of CO_2 and H_2SO_4 (C = 12, O = 16, H = 1, S = 32). — 3

18. Write the formula of the compounds from the ions: — 3
(a) Na^+ and SO_4^{2-}
(b) Ca^{2+} and OH^-.

OR

Explain why helium exists as single atoms while oxygen exists as O_2. — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between ionic and covalent compounds on bond, state, melting point, conductivity and solubility. — 5

OR

Explain atoms, molecules, ions and formula units with examples, and the difference between atomicity and valency. — 5

20. Explain the rules for writing chemical formulae with the examples of Al_2O_3, $CaCO_3$ and $(NH_4)_2SO_4$. — 5

OR

The percentage of calcium in calcium carbonate is to be found. Calculate it and the mass of calcium in 250 g of limestone (CaCO3, Ca = 40, C = 12, O = 16). — 5

END OF QUESTION PAPER — Total: 40 Marks

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